Kush Gupta

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LinkedIn | GitHub | Portfolio



Summary

- B.Tech Computer Science student seeking a **Software Development** role, with hands-on experience building and deploying full-stack web applications using **React.js, Next.js, TypeScript, and Node.js**. Experienced in designing **RESTful APIs**, integrating **AI (Google Gemini)**, and building **real-time systems** with WebSockets and WebRTC. Strong problem solver with **750+ DSA problems** solved and a **LeetCode rating of 1861**.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Information Technology, Una	7.60 (Current)	2023–Present
Senior Secondary	CBSE Board	93.2%	2023
Secondary	CBSE Board	99%	2021

Projects

Career Wave **Aug. 2026 - Sept. 2026**
Next.js, TypeScript, Prisma, PostgreSQL GitHub | Live Demo

- **Architected a Full-Stack ATS Platform:** Built an enterprise job board with **Next.js 15 App Router**, **TypeScript**, and **Prisma ORM** on serverless **PostgreSQL (Neon)**, replacing separate REST APIs with type-safe **Server Actions** and **React Server Components**.
- **Engineered Gemini AI ATS Evaluator:** Integrated **Google Gemini 3.8 Flash** to compute real-time resume-to-job **ATS match scores**, highlight candidate strengths, and identify **keyword gaps**, alongside a conversational career discovery assistant.
- **Integrated Stripe Monetization & Inngest Scheduling:** Monetized job postings via **Stripe Checkout** with tiered listing durations and **webhook-verified** activation; eliminated traditional cron servers by scheduling automated post expirations with **Inngest** serverless events.
- **Built Recruiter ATS Dashboard & Notifications:** Delivered a recruiter management suite with **candidate pipeline** status tracking, an in-browser **PDF resume previewer**, candidate application withdrawal, and **Resend** transactional email alerts for application milestones.
- **Hardened Edge Security & Cloud Storage:** Implemented **NextAuth v5** role-based onboarding (Recruiter vs. Seeker), guarded mutations with **Arcjet** bot shielding and **Zod** validation, and integrated **UploadThing** for secure cloud resume and logo storage.

SketchBattle **Mar. 2026 – May 2026**
React, TypeScript, Node.js, Socket.IO, MongoDB GitHub | Live Demo

- **Real-Time Multiplayer Canvas:** Built a synchronized drawing engine with **sub-50ms latency** using **React 18**, **TypeScript**, and **Socket.IO**, featuring a **4-way BFS flood fill** algorithm, stroke undo stack, and mid-round canvas history replay upon reconnection.
- **Algorithmic Game Engine:** Implemented core mechanics including speed-tiered rank scoring, **Levenshtein distance** near-miss typo alerts ("Very Close!"), progressive word hints, and customizable room settings with difficulty tiers.
- **P2P Video & Synthesized Audio:** Integrated room-scoped **WebRTC** peer-to-peer video streaming alongside **WebSockets**, and developed a zero-dependency sound effects synthesizer using the native browser **Web Audio API**.
- **Mobile-First UX & Voice Input:** Designed a fully **responsive** party-game UI featuring a dedicated main-screen live chat guess input for mobile users, complemented by **Web Speech API** voice recognition for hands-free guessing.
- **Security, Persistence & CI/CD:** Hardened the backend with **bcrypt** password hashing, sliding-window **IP rate limiting**, and 24-hr **MongoDB TTL indexes**; authored **15 automated tests** integrated into a **GitHub Actions CI** pipeline.

Streamify **Apr. 2025 – June 2025**
React, Node.js, Express.js, MongoDB, Stream SDK GitHub | Live Demo

- **Smart Reciprocal Matching:** Engineered a **mutual language-scoring algorithm** categorizing partners into tiered compatibility ratings (100% Perfect Exchange, Native Speaker, Study Partner) with real-time search and language filters.

- **Google Gemini AI Assistant:** Integrated **Gemini 3.8 Flash** in chat with **structured JSON output** for bi-directional translation with phonetics, grammar diagnostics with explanations, and contextual conversation icebreakers.
- **Real-Time Chat & WebRTC Video:** Implemented low-latency messaging and **video calling** via **Stream SDKs**, enforcing backend **room-level authorization** checks (`/api/chat/verify-call`) and in-chat call notification prompts.
- **Security & Database Optimization:** Hardened authentication using **httpOnly JWT cookies**, **bcrypt**, and **rate limiting**; optimized social queries (send, accept, reject, unfriend) with **compound MongoDB indexes**.
- **Cloud Media & Automated Testing:** Built a **Cloudinary CDN** media pipeline with automated face-centering crops and 5MB validation; ensured reliability with automated test suites using **Vitest** and **Node.js test runner**.

Skills

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js (App Router, Server Actions, RSC), HTML, CSS, Tailwind CSS, Redux, Zustand, jQuery

Backend: Node.js, Express.js, REST APIs, Socket.IO, WebSockets, JWT Authentication, NextAuth, Zod, Arcjet

Databases & ORM: MongoDB, PostgreSQL (Neon), Prisma ORM, Firebase

AI & Real-Time: Google Gemini API, WebRTC, Stream SDK, Web Audio API, Web Speech API

Integrations & Services: Stripe, Inngest, Resend, UploadThing, Cloudinary, Clerk

Testing & DevOps: Vitest, Node.js Test Runner, GitHub Actions (CI/CD), Git, GitHub, Docker, AWS, Vercel, Render

Libraries: NumPy, Pandas, Matplotlib, Seaborn

Tools: VS Code, Replit, LaTeX

Soft Skills: Problem Solving, Analytical Thinking, Leadership (Chess Captain), Teamwork & Collaboration, Communication, Time Management, Adaptability, Quick Learning

Key Courses Taken

Mathematics: Linear Algebra, Basic Calculus, Discrete Mathematics, Probability & Random Processes

Core Subjects: DSA, OOPS, Discrete Structures, Computer Organization, Operating Systems, Digital System Design, Computer Networks, DBMS, System Design, Compiler Design, Parallel Architecture and Processing, Machine Learning, Deep Learning, Software Development

Achievements

- Solved **750+ DSA problems** across **LeetCode**, **GeeksforGeeks**, and other platforms; **Knight** on LeetCode with a contest rating of **1861**.
- Completed AlgoUniversity's **Dynamic Programming Camp** led by **Codeforces Master** Manas Kumar Verma, covering **Matrix Exponentiation** and **SQRT Decomposition**.
- **FIDE-rated** chess player (**FIDE ID: 25129937**) with **national and international** tournament experience; served as **chess captain** at Inter IIIT and secured **2nd place** in the online Inter IIIT tournament (2025).